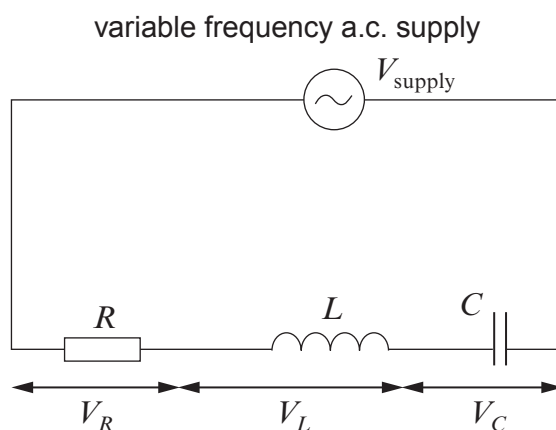


## Option A – Alternating Currents

6. (a) The following *RCL* circuit is constructed.



For this *RCL* circuit, describe the relationships between the rms pds  $V_{\text{supply}}$ ,  $V_R$ ,  $V_L$  and  $V_C$ :

- (i) when the circuit is not in resonance;  
*Space for diagram.*

[3]

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- (ii) when the circuit is in resonance.

[2]

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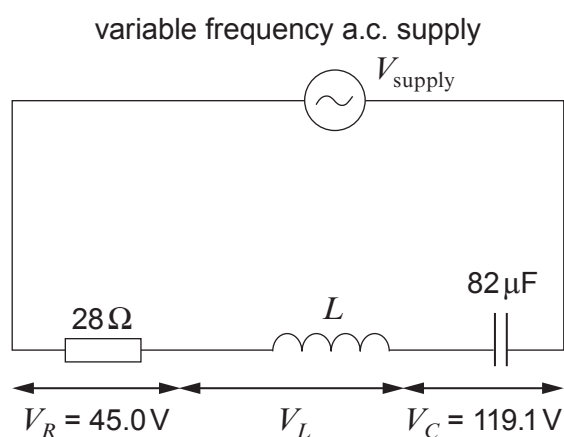
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- (b) The following *RCL* circuit is **at resonance**. The values of *R* and *C* are provided along with their rms pd values.



Calculate:

- (i) the *Q* factor of the circuit; [1]

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- (ii) the rms current; [1]

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- (iii) the frequency of the power supply; [2]

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- (iv) the inductance of the inductor. [2]

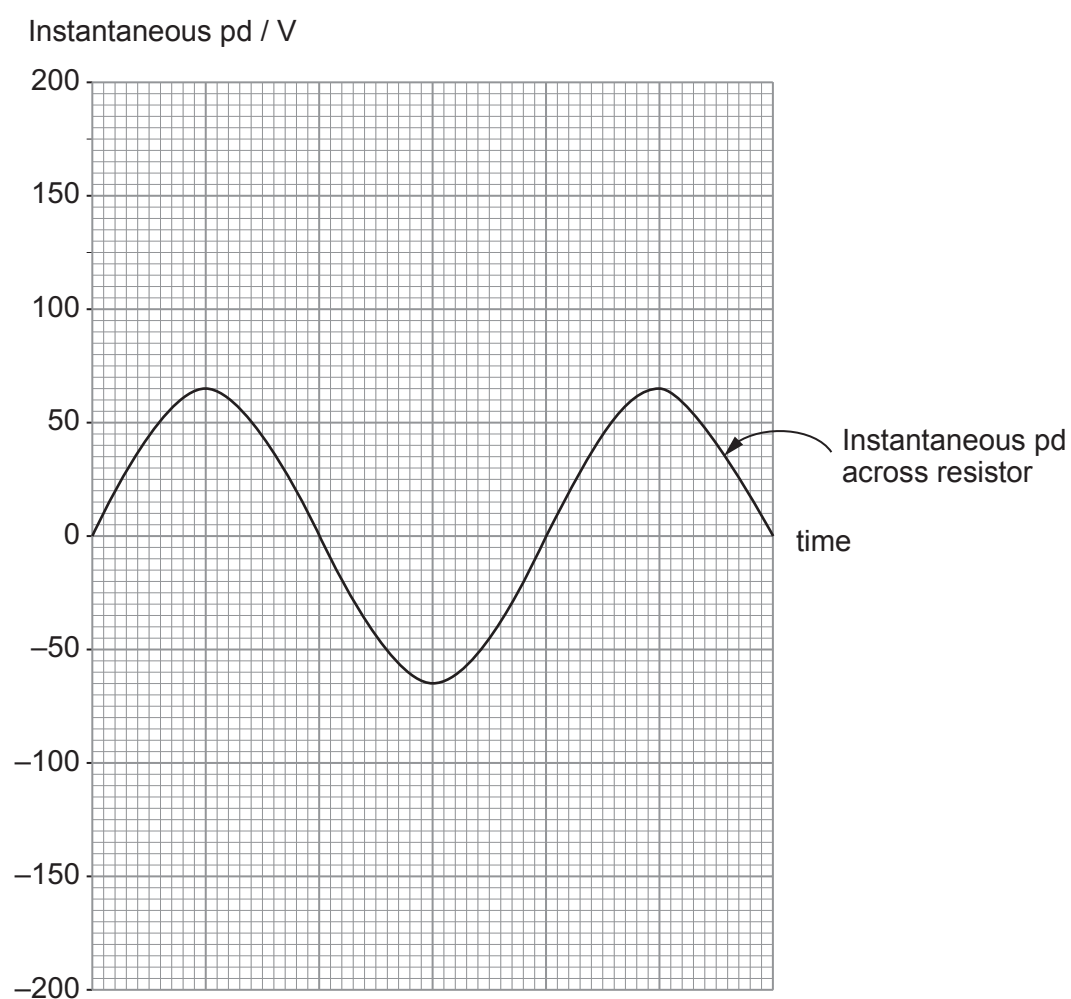
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- (v) Sketch a graph of the instantaneous pd across the capacitor on the grid provided. (The instantaneous pd across the resistor is shown.) [2]



- (vi) Without further calculation, explain why the current decreases when the frequency of the power supply is increased. [2]

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- (c) Sam claims that the output pd ( $V_{OUT}$ ) in the following circuit is greater than 6.0 V when the frequency is greater than 20 kHz, but less than 6.0 V when below 20 kHz. Investigate whether or not Sam is correct. [5]

